FMTH0306/Rev.1.0

Course Plan

Semester: V Year: 2026-27

Course Title: Cryptography and Network Security	Course Code: **26ECAC305**
Total Contact Hours: 68 Hrs	Duration of ESA: 3 Hrs
ISA Marks: 50	ESA Marks: 50
Lesson Plan Authors: Prof. Pooja Shettar	Date: 07-08-2026
Checked By: Dr. Narayan D G	Date: 14-08-2026

Prerequisites:

This course requires basic knowledge of discrete mathematics and number theory (modular arithmetic, prime numbers), elementary probability, and the fundamentals of computer networks and the OSI/TCP-IP protocol stack.

Course Outcomes (CO'S):

At the end of the course, the student should be able to:

1.Illustrate cryptographic algorithms and key management techniques to provide secure data communication.
2.Analyze authentication and authorization algorithms to ensure data integrity, authentication, and non-repudiation.
3.Apply transport, network, application layer protocols to establish secure communication.
4.Evaluate post-quantum and lightweight cryptographic techniques for securing next generation networks.

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (POs)

Course Title: Cryptography and Network Security	Semester: V
Course Code: **26ECAC305**	Year: 2026-27

Course Outcomes (COs) / Program Outcomes (POs)	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	
1.Illustrate cryptographic algorithms and key management techniques to provide secure data communication.	H	H			M							M		
2.Analyze authentication and authorization algorithms to ensure data integrity, authentication, and non-repudiation.		H			M							M		
3.Apply transport, network, application layer protocols to establish secure communication.	H	H			M							M		
4.Evaluate post-quantum and lightweight cryptographic techniques for securing next generation networks.		H			M							M		

Degree of compliance **L**: Low **M**: Medium **H**: High

Competency addressed in the Course, corresponding Performance Indicators (PI) and Course Specific Performance Indicators (CSPI)

Competency	Performance Indicators (PI)	Course Specific Performance Indicators (CSPI)
1.1 - Demonstrate competence in mathematical modeling.	1.1.1 - Apply the knowledge of discrete structures, number theory and numerical techniques to solve problems.	1.1.1.1 - Apply modular arithmetic, number theory (prime numbers, GCD, Euler's totient) and numerical techniques to solve cryptographic problems such as classical ciphers, RSA, Diffie–Hellman key exchange and ECC computations.
2.1 - Demonstrate an ability to identify and characterize an engineering problem.	2.1.3 - Identifies mathematical/algorithmic knowledge that applies to a given problem. 2.1.4 - Use processes/techniques for problem analysis.	2.1.3.1 - Identify and select appropriate cryptographic algorithms (symmetric, asymmetric, hash, digital signature, lightweight/post-quantum) suited to a given security requirement. 2.1.4.1 - Analyze the security requirements of a given communication/networking scenario to determine appropriate authentication, authorization and protocol-level solutions.
5.2- Demonstrate an ability to select and apply domain-specific tools, modelling techniques, and resources	5.2.1- Select suitable mathematical, statistical, or analytical tools based on the nature and complexity of the engineering problem, data, or modelling requirements	5.2.1.1 Demonstrate security analysis tools such as OpenSSL, Wireshark, CrypTool,iptables, Snort, PKI tools, JWT/OAuth implementations and PQC tools/libraries.
12.1- Demonstrate an ability to develop robust, scalable, and maintainable software systems using appropriate design principles.	12.1.3-Integrate third-party libraries, APIs, and frameworks to enhance functionality.	12.1.3.1Implement secure communication and data protection mechanisms by integrating cryptographic libraries, security APIs, and network security tools into software applications

Course Content

Course Code: **26ECAC305**	Course Title: Cryptography and Network Security	
L-T-P : 3-0-1	Credits: 4	Contact Hrs. 5Hrs/week
ISA Marks: 50	ESA Marks: 50	Total Marks: 100
Teaching Hrs.: 40	Practical Hrs: 28	Exam Duration: 3Hrs

Content	Hrs
Unit - 1	
Chapter 1 **Introduction** Computer Security Principles, The OSI Security architecture: Security attacks, services and mechanisms, Threat model for Network Security, Classical Encryption techniques: Substitution ciphers: Caesar, Monoalphabetic, Playfair and Hill ciphers, Transposition ciphers, Taxonomy of Cryptography and Cryptanalysis	**4 hrs**
Chapter 2 **Symmetric Key Cryptography** Overview of Symmetric Key Cryptography, Stream Ciphers: RC4, Block Ciphers, Data Encryption Standard (DES), Advanced Encryption Standard (AES), Block Cipher Modes of Operation, Authenticated Encryption and AEAD, AES-GCM - Overview and Applications.	**6 hrs**
Chapter 3 **Asymmetric Cryptography and Key Management**: Overview of Asymmetric Key Cryptography, Diffie-Hellman Key Exchange, RSA, ElGamal Cryptosystem, Elliptic Curve Cryptography (ECC), Symmetric Key Distribution, Public Key Distribution, Digital Certificates, X.509 Certificates, Public Key Infrastructure.	**6 hrs**
Unit - 2	
Chapter 4 **Data Integrity Algorithms** Cryptographic Hash Functions, SHA-2 (SHA-512), SHA-3 (Keccak) – overview, and applications of cryptographic hash functions, Message Authentication Codes (MAC, HMAC), Digital Signatures: RSA, ECDSA. Applications of Digital Signatures	**6 hrs**

Chapter 5 **Authentication & Authorization:** Authentication methods: Passwords, Two-Factor and Multi-Factor Authentication, Single Sign-On (SSO), Authentication protocols, Zero Knowledge Proofs, Authorization models: Access Control Matrix, Role-Based Access Control, Firewalls and Intrusion Detection Systems	**4 hrs**
Chapter 6 **Application and Transport Security Protocols**: Secure email: Pretty Good Privacy (PGP), S/MIME, Transport security: SSL, TLS, Secure Shell, Kerberos authentication system, API security: token-based authentication, JSON Web Tokens (JWT), OAuth	**6 hrs**
Unit - 3	
Chapter 7 **Network and Wireless Security Protocols**: IPSec overview, Encapsulating Security Payload, Internet Key Exchange, Wireless security: IEEE 802.11, modern wireless security protocols (WPA2/WPA3),5G network security architecture, secure communication in distributed systems.	**4 hrs**
Chapter 8 **Emerging Cryptography**: Post-Quantum Cryptography: Post-Quantum Cryptographic (PQC) Algorithms, Classical cryptography vs. PQC, PQC in Network Security, Lightweight Cryptography in Emerging Networks: Design considerations, Lightweight cryptographic algorithms, Applications in IoT and edge networks.	**4 hrs**

Text Books

1. William Stallings, Cryptography and Network Security: Principles and Practices, 8th Edition, Pearson, 2024.

2. Mark Stamp, Information Security: Principles and Practices, 3rd Edition, John Wiley and Sons, 2021.

References

1. Jonathan Katz and Yehuda Lindell, Introduction to Modern Cryptography, 3rd Edition, CRC Press, 2020.

2. S. Musa, Network Security and Cryptography, 1st Edition, Springer, 2022.

Evaluation Scheme

ISA Scheme

Assessment	QP Max Marks	Weightage in Marks
In Semester Assessment -1	40	33
In Semester Assessment -2	40	
Programming Assessments	20	17
	TOTAL	50

Course Unitization for ISA and ESA

Topics / Chapters	Teaching Hours	No. of Questions in ISA-1	No. of Questions in ISA-2	No. of Questions in Activity	No. of Questions in ESA
Unit I					
1. Introduction	04	1	0		1
2. Symmetric Key Cryptography	06	1.5	0	1	1.5
3. Asymmetric Cryptography and Key Management	06	1.5	0	1	1.5
Unit II					
4. Data Integrity Algorithms	06	0	1.5	1	1.5
5. Authentication and Authorization	04	0	1	1	1
6. Application and Transport Security Protocols	06	0	1.5	1	1.5
Unit III					
7. Network and Wireless Security Protocols	04	0	0	1	1
8. Emerging Cryptography	04	0	0	1	1

Note

1. Each Question carries 20 marks and may have sub-questions.
2. Mixing of sub-questions from different chapters within a unit (only for Unit I and Unit II) is allowed in ISA I, II and ESA
3. Answer 5 full questions of 20 marks each (two full questions from Unit I, II and one full question from Unit III) out of 8 questions in ESA.

Course Assessment Plan

Course Title: Cryptography and Network Security Code: 26ECAC305					
Course outcomes (COs)	**Weightage in assessment**	**Assessment Methods**			
		ISA – I	**ISA - II**	**Assignment**	**ESA**
1.Illustrate cryptographic algorithms and key management techniques to provide secure data communication.	25%	✓		✓	✓
2.Analyze authentication and authorization algorithms to ensure data integrity, authentication, and non-repudiation.	25%	✓		✓	✓
3.Apply transport, network, application layer protocols to establish secure communication.	25%		✓	✓	✓
4.Evaluate post-quantum and lightweight cryptographic techniques for securing next generation networks.	25%		✓	✓	✓
	Weightage	16%	17%	17%	50%

Lab Activity

SL.No.	Activity Title	Slots (1 slot =2 hrs)
1	Performance and Security Analysis of Symmetric Key Algorithms	01
2	Performance and Security Analysis of Asymmetric Key Algorithms	01
3	Performance and Security Analysis of Cryptographic Hash Functions	01
4	Performance and Security Analysis of Digital Signature Algorithms	01
	Evaluation - 1	01
5	TLS handshake analysis using Wireshark	01
6	PKI demonstration - generating a CA, issuing and verifying certificates, and building a chain of trust	02
7	Firewall rule configuration and traffic filtering using iptables	01
	Evaluation -2	01
8	Intrusion detection using Snort — writing rules and analyzing alerts on simulated attack traffic	01
9	JWT-based authentication and OAuth 2.0 authorization flow demonstration	01
10	Implement post-quantum key establishment and post-quantum digital signatures, and analyze their security and performance	01
	Evaluation - 3	01
	Total	28 Hrs

Assessment parameters and Rubrics for Course activity

Parameter	Excellent (8–10)	Good (6–7)	Satisfactory (4–5)	Needs Improvement (0–3)
Program Design	Complete, modular, well-structured, and optimized	Well-structured with few design inconsistencies	Basic implementation with limited modularity	Incomplete, poorly structured, or incorrect implementation
Execution	Executes correctly for all test cases	Executes correctly with few deviations	Executes only for selected or simple test cases	Does not execute or produces incorrect results
Conceptual Understanding	Explains algorithms and security concepts confidently with clear justification	Demonstrates good conceptual clarity with adequate explanations	Shows basic understanding with limited explanations	Unable to explain concepts or justify the implementation
Testing & Analysis	Uses multiple test cases, validates outputs, and analyzes results effectively	Tests with adequate test cases and provides basic analysis	Performs limited validation with inadequate analysis	No testing or analysis performed
Documentation	Complete, clear, well-organized, and professionally presented	Well-organized with sufficient details	Basic documentation covering essential aspects	Incomplete, or poorly organized documentation

Date : **HoD CSE-AI**

Chapter wise Plan

Course Code and Title: **26ECAC305** Cryptography and Network Security	
Chapter Number and Title: 1. Introduction	Planned Hours: **4 hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Sl.No	TLO's	CO's	BL	CA Code
1.	Describe the OSI security architecture	CO1	L1	1.1
2	Classify security attacks, services and mechanisms	CO1	L2	1.1
3.	Explain the working of classical substitution and transposition ciphers	CO1	L2	1.1
4.	Apply classical cipher techniques to encrypt and decrypt simple messages	CO1	L3	2.1

Lesson Schedule Class No. - Portion covered per hour
1. Computer Security Principles, OSI Security architecture: Security attacks, services and mechanisms
2. A model for Network Security
3. Classical Encryption techniques: Substitution ciphers – Caesar, Monoalphabetic, Playfair and Hill ciphers
4. Transposition ciphers, Taxonomy of Cryptography and Cryptanalysis

Review Questions

Sr.No. - Questions	TLO	BL	PI Code
1. List and briefly explain the different categories of security attacks and services.	1	L2	1.1.1
2. With a neat diagram, explain the OSI security architecture and a general model for network security.	1	L2	1.1.1
3. Encrypt the message “COMPUTER SCIENCE” using the Playfair cipher with the keyword “SECURITY”.	3	L2	1.1.1
4. Explain the working of the Hill cipher. Encrypt the message “PAY” using the key matrix [[9,4],[5,7]] (mod 26).	3	L2	1.1.1
5. What is a transposition cipher? Explain the working of a columnar transposition cipher with the keyword “SECURITY” applied to the message “CRYPTOGRAPHY”.	4	L2	1.1.1
6. A college network observes that an attacker is secretly reading students' login credentials while they are being transmitted.a) Identify the attack. b) Identify the security property affected. c) Suggest one suitable security mechanism.	2	L3	2.1.3

7. Encrypt the message We are all together Using double transposition cipher with 4 rows and 4 columns, using row permutation (1,2,3,4), (2,4,1,3) and column permutation (1,2,3,4) à (3,1,2,4) Assign each letter of the same message given, to a unique symbol and then perform double transposition same as above problem. Which method is more secure? Discuss advantages and disadvantages of each method	3	L3	2.1.3
8.Find the plaintext and the key from cipher text CSYEVIXIVQMREXIH given that the cipher is simple substitution of the shift-by-n variety	3	L2	1.1.1
9 .Use the keyword “INFORMATIONSECURITY” with playfair algorithm and decrypt the message UYSBCORMLXBDUT. How secure is a playfair cipher compared to mono alphabetic cipher?	3	L2	1.1.1
10.A company uses the following 3×3 Hill cipher key matrix to encrypt a message: a)Verify whether the given matrix is a valid Hill cipher key modulo 26. b)Encrypt the plaintext ACTNOW using the given key matrix. c)Write the resulting cipher text. 6 24 1 13 16 10 20 17 15	3	L3	2.1.3
11. An attacker observes that the letter Q occurs very frequently in a long ciphertext. The attacker suspects that Q represents E. a) Which cryptanalysis technique is being used? b) Why can this technique work against monoalphabetic ciphers? c) Mention one limitation of this technique.	3	L3	2.1.4
12.A company encrypts its confidential messages using a Caesar cipher. An attacker intercepts several messages and determines the key by trying all possible shifts. a) Identify the cryptanalysis technique. b)Why is Caesar cipher vulnerable to this attack? c)Suggest a more secure classical cipher from the syllabus.	4	L3	2.1.4

Chapter wise Plan

Course Code and Title: **26ECAC305** Cryptography and Network Security	
Chapter Number and Title: : 2. Symmetric Key Cryptography	Planned Hours: **6 hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Sr.No	TLO's	CO's	BL	CA Code
1.	Explain the working principle of stream ciphers with reference to A5/1 and RC4	CO2	L2	1.1
2.	Describe the structure and design of the DES and AES algorithms	CO2	L2	1.1
3.	Apply block cipher modes of operation to encrypt data and analyze their strengths and weaknesses	CO2	L3	2.1

Lesson Schedule Class No. - Portion covered per hour
1. Overview of Symmetric Key Cryptography
2. Stream Ciphers: A5/1 and RC4
3. Block Ciphers – structure and design principles
4. Data Encryption Standard (DES)
5. Advanced Encryption Standard (AES)
6. Block Cipher Modes of Operation

Review Questions

Sr.No. - Questions	TLO	BL	PI Code
1.Explain the working of the A5/1 stream cipher used in GSM communication.	1	L2	1.1.1
2. Explain the working of the RC4 stream cipher with a neat diagram.	1	L2	1.1.1
3. With a neat diagram, explain the design of the DES algorithm and discuss the significance of the S-boxes.	2	L2	1.1.1
4. Explain the structure of AES. Why is AES considered more secure than DES?	2	L2	1.1.1
5. Differentiate between ECB, CBC, CFB and OFB modes of operation. Discuss one advantage and one limitation of each.	3	L2	1.1.1

6. A sender and receiver use the same secret key for secure communication. Explain the working of symmetric key encryption and analyze two challenges that arise when the number of communicating users increases.	1	L3	2.1.3
7. Differentiate between AES and DES. With a neat diagram explain the design of the DES algorithm. Analyze security of DES	2	L2	1.1.1
8. Draw the structure of one DES round and explain how the **Expansion, XOR with subkey, S-box, and P-box** operations transform the 32-bit right half.	2	L2	1.1.1
9. For a simplified RC4 system, the initial permutation is S = [0, 1, 2, 3] and the key is K = [2, 1, 3, 0]. Perform the KSA for the first four iterations and obtain the resulting S-box.	1	L3	2.1.3
10. A block cipher uses a 64-bit block size and a 128-bit key. If a file contains 150 bytes, determine the number of complete blocks and the number of padding bytes required. Justify the significance of padding.	2	L3	2.1.3
11. Consider an AES state containing the following first row: **[19, A0, 9A, E9]**. Explain how **ShiftRows** transforms the state and describe how **MixColumns** contribute to diffusion.	2	L3	2.1.3
12. A 128-bit block cipher is used to encrypt a 1000-byte file using ECB mode. Determine the number of cipher blocks required, including padding, and analyze why ECB may reveal patterns in the plaintext	3	L3	2.1.3
13. A system encrypts a message but does not provide authentication. An attacker modifies some ciphertext bits before delivery. Analyze what security problem occurs and explain how authenticated encryption prevents such modification.	1	L3	2.1.4
14. An application sends 100 bytes of confidential data **along with** 20 bytes of information that must remain visible but must be authenticated**.** Explain how AEAD handles the two types of data and identify which part is encrypted and which part is authenticated only.	1	L3	2.1.4
15. A network application uses AES-GCM with a 96-bit nonce and a 128-bit authentication tag. Explain why nonce uniqueness is critical and analyze the security consequence of reusing the same nonce with the same key.	2	L3	2.1.3

Chapter wise Plan

Course Code and Title: **26ECAC305** Cryptography and Network Security	
Chapter Number and Title: 3. Asymmetric Cryptography and Key Management	Planned Hours: **6 hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Sr.No	TLO's	CO's	BL	CA Code
1.	Demonstrate the use of public-key cryptosystems for secure communication in a given scenario.	CO2	L3	1.1
2.	Solve problems using the Diffie-Hellman key exchange, RSA and ElGamal cryptosystem	CO2	L3	2.1
3.	Apply the ECC key generation and encryption/decryption process for a given set of elliptic curve parameters.	CO2	L3	1.1
4.	Analyze key distribution techniques, digital certificates, X.509 and the role of a Public Key Infrastructure	CO2	L3	2.1

Lesson Schedule Class No. - Portion covered per hour
1. Overview of Asymmetric Key Cryptography, Diffie–Hellman Key Exchange
2. RSA Algorithm
3. ElGamal Cryptosystem
4. Elliptic Curve Cryptography (ECC)
5. Symmetric Key Distribution and Public Key Distribution
6. Digital Certificates, X.509 Certificates and Public Key Infrastructure

Review Questions

Sr.No. - Questions	TLO	BL	PI Code
1.Discuss the man-in-the-middle attack on Diffie–Hellman key exchange and how it can be mitigated.	2	L2	1.1.1
2.. Discuss the security of RSA and significance of the integer factorization problem.	1	L2	1.1.1
3. Compare RSA and ElGamal cryptosystems in terms of security and efficiency.	1	L2	1.1.1
4. Explain the mathematical basis of Elliptic Curve Cryptography.	3	L2	1.1.1
5. Explain public key distribution methods including public announcement and publicly available directories.	4	L2	1.1.1
6. Explain the structure of an X.509 certificate and describe how a Public Key Infrastructure supports secure key distribution.	4	L2	1.1.1
7. Describe the role of a Certificate Authority (CA) and the process of certificate issuance and verification.	4	L2	1.1.1
8. Explain the RSA algorithm with key generation, encryption and decryption steps, illustrated with a small example.	2	L3	2.1.3
9.Explain the ElGamal cryptosystem with an example, showing key generation, encryption and decryption.	2	L3	2.1.3
10. Illustrate point addition and point doubling on an elliptic curve with a suitable example.	3	L2	1.1.1
11. Alice and Bob want to establish a common secret key over an insecure network. Given q=23, primitive root α=5, Alice's private key X_A=6, and Bob's private key X_B=15, apply the Diffie-Hellman algorithm to compute their public keys and the shared secret key. Verify that both users obtain the same K.	3	L3	2.1.3
12. A company wants to transmit a confidential numeric message using RSA. Given p=11, q=17, and e=7, generate the RSA public and private keys and apply them to encrypt and decrypt the message M=8. Verify the correctness of the recovered message.	3	L3	2.1.3

13.A secure messaging application uses ElGamal encryption. Given prime p=23, generator g=5, receiver's private key x=6, message M=10, and random integer k=3, apply the ElGamal cryptosystem to generate the public key and determine the ciphertext. Show how the receiver can recover the original message.	3	L3	2.1.3
14. A battery-powered IoT sensor has limited processing capability, memory, and communication bandwidth but still requires strong public-key security. Apply the characteristics of Elliptic Curve Cryptography to explain why ECC would be suitable for this system compared with conventional public-key approaches using much larger keys.	3	L3	2.1.4
15.A company has 100 employees who need to communicate securely using symmetric encryption. Directly distributing a separate secret key for every pair of employees becomes difficult. Apply an appropriate symmetric-key distribution mechanism using a Key Distribution Center (KDC) and illustrate how two employees can securely obtain a session key.	3	L3	2.1.4

Chapter wise Plan

Course Code and Title: **26ECAC305** Cryptography and Network Security	
Chapter Number and Title: 4. Data Integrity Algorithms	Planned Hours: **6 hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Sr.No	TLO's	CO's	BL	CA Code
1	Use cryptographic hash functions to verify the integrity of a given message.	CO3	L3	1.1
2	Compute the SHA-512 hash value for a given message using the specified processing steps.	CO3	L3	1.1
3	Construct a MAC and HMAC for a given message and secret key.	CO3	L3	2.1
4	Generate and verify digital signatures using RSA and ECDSA for a given message and parameters.	CO3	L3	1.1

Lesson Schedule Class No. - Portion covered per hour
1. Cryptographic Hash Functions: requirements and applications
2. SHA-512
3. Message Authentication Codes (MAC) and HMAC
4. Digital Signatures: RSA-based Digital Signature
5. Digital Signatures: ECDSA
6. Applications of Hashing and Digital Signatures

Review Questions

Sr.No. - Questions	TLO	BL	PI Code
1. List the requirements a cryptographic hash function must satisfy. Discuss why collision resistance is important.	1	L2	1.1.1
2. With a neat diagram, explain the structure of the SHA-512 algorithm.	2	L2	1.1.1
3. Explain the compression function used in SHA-512.	2	L2	1.1.1

4. Explain the design of the SHA-3 (Keccak) algorithm and the sponge construction.	2	L2	1.1.1
5. Given a message and a shared secret key, demonstrate the generation and verification of an HMAC and explain how it provides message authentication.	3	L3	2.1.3
6. Apply the RSA digital signature scheme to demonstrate the signing and verification of a message using suitable numerical values.	3	L3	2.1.3
7. A software vendor distributes executable files over the Internet. Apply cryptographic hashing and digital signatures to design a mechanism that allows users to verify both integrity and authenticity of the downloaded files.	3	L3	2.1.3
8. Differentiate between a MAC and a digital signature.	3	L2	1.1.1
9. Explain the birthday attack on hash functions and its implications for hash output length.	1	L2	1.1.1
10. A file is transmitted from a sender to a receiver. Demonstrate how a cryptographic hash function can be used to verify whether the file has been modified during transmission..	2	L3	2.1.3
11. Explain the working of the Elliptic Curve Digital Signature Algorithm (ECDSA).	4	L2	1.1.1
12. Compare RSA-based digital signatures with ECDSA in terms of key size and performance.	4	L2	1.1.1
13. Discuss the applications of digital signatures in software distribution, legal documents and blockchain.	4	L3	2.1.3
14. What is a message digest? Explain how it ensures data integrity during transmission.	1	L1	1.1.1
15. Apply RSA-based digital signatures to an electronic certificate system and demonstrate how the receiver verifies the authenticity and integrity of the certificate.	4	L4	2.1.3
16. Explain how digital signatures provide non-repudiation in secure communication.	4	L2	1.1.1

Chapter wise Plan

Course Code and Title: 2**26ECAC305** Cryptography and Network Security	
Chapter Number and Title: 5. **Authentication & Authorization**	Planned Hours: **4 hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Sr.No	TLO's	CO's	BL	CA Code
1	Implement appropriate authentication methods to secure user access in a given system scenario.	CO2	L3	2.1
2	Demonstrate the use of SSO to provide authenticated access to multiple applications.	CO2	L2	2.1
3	Apply suitable authentication protocols to establish secure communication between users and network services.	CO2	L3	2.1
4	Apply RBAC to assign appropriate permissions to users based on a given organizational scenario.	CO2	L3	2.1
5	Apply firewalls, IDS, cloud access control and Zero Trust Architecture to secure a given system or network scenario.	CO2	L3	2.1

Lesson Schedule Class No. - Portion covered per hour
1. Authentication Methods: Password-based authentication, Two-Factor Authentication (2FA), Multi-Factor Authentication (MFA)
2. Single Sign-On (SSO) and Authentication Protocols: Working of SSO and overview of common authentication protocols
3. Authorization Models: Access Control Matrix and Role-Based Access Control (RBAC) with examples
4. Network and System Security: Firewalls, Intrusion Detection Systems (IDS), Cloud Access Control and Zero Trust Architecture

Review Questions

Sr.No. - Questions	TLO	BL	PI Code
1. Explain the different authentication methods, including passwords, Two-Factor Authentication (2FA), and Multi-Factor Authentication (MFA).	1	L2	2.1.3

2. A university provides students access to an online examination portal. Apply an appropriate authentication method using passwords and 2FA, and describe the authentication process.	1	L3	2.1.4
3. Explain the working of Single Sign-On (SSO) and describe its advantages in an enterprise environment.	2	L2	1.1.1
4. An organization has multiple applications that employees access using separate usernames and passwords. Apply the concept of SSO to this scenario and illustrate the authentication flow.	2	L3	2.1.4
5. Describe the working of common authentication protocols and explain their role in secure user authentication.	3	L2	1.1.1
6. A university has a centralized authentication system for students and faculty. Apply an appropriate authentication protocol to the scenario and explain how a user can securely authenticate and access a protected service	3	L3	2.1.4
7. Explain the Access Control Matrix and Role-Based Access Control (RBAC) models with suitable examples.	4	L2	2.1.3
8. Consider a company with three roles—Administrator, Faculty, and Student—and three resources—Student Records, Course Materials, and Examination System. Construct an Access Control Matrix specifying suitable access permissions for each role.	4	L3	2.1.4
9. Explain the role of firewalls and Intrusion Detection Systems (IDS) in protecting a computer network.	5	L2	1.1.1
10. A web server is receiving repeated unauthorized connection attempts from a particular IP address. Apply suitable firewall rules and IDS mechanisms to detect and restrict the suspicious traffic.	5	L3	2.1.4

Chapter wise Plan

Course Code and Title: **26ECAC305** Cryptography and Network Security	
Chapter Number and Title: 6. **Application and Transport Security Protocols**	Planned Hours: **6 hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Sr.No	TLO's	CO's	BL	CA Code
1	Implement PGP and S/MIME mechanisms for securing email communication in a given scenario.	CO3	L3	2.1
2	Demonstrate the use of SSL/TLS to establish secure communication between clients and servers in a given scenario.	CO3	L2	1.1
3	Configure SSH and Kerberos authentication mechanisms for secure access to network resources in a given scenario.	CO3	L3	2.1
4	Apply token-based authentication mechanisms to secure access to web applications and services.	CO3	L3	2.1
5	Implement appropriate application and transport security protocols to establish secure communication for a given scenario.	CO3	L3	2.1

Lesson Schedule Class No. - Portion covered per hour
1. Secure Email: Pretty Good Privacy (PGP) – concepts, architecture and working
2. Secure Email: S/MIME – concepts, working and security services; comparison of PGP and S/MIME
3. Transport Security: SSL – concepts, architecture and working
4. Transport Security: TLS – architecture, TLS handshake procedure and secure session establishment
5. Secure Remote Access and Authentication: Secure Shell (SSH) and Kerberos authentication system
6. API Security: Token-based authentication, JSON Web Tokens (JWT) and OAuth authorization flow

Review Questions

Sr.No. - Questions	TLO	BL	PI Code
1. Explain the working of Pretty Good Privacy (PGP) and describe how it provides secure email communication. .	1	L2	1.1.1
2. Explain the security services provided by S/MIME for protecting email communication.	2	L2	1.1.1
3. An organization wants to send confidential emails containing sensitive business information. Apply PGP to the given scenario and describe the steps involved in securing and sending the email.	1	L3	2.1.4
4. Explain the SSL/TLS handshake procedure with a neat diagram.	2	L2	1.1.1
5. A company wants to secure communication between a web browser and its online banking server. Apply the SSL/TLS protocol and describe the major steps involved in establishing a secure session.	2	L3	2.1.4
6. Explain the working of the Kerberos authentication system and describe the role of the Authentication Server and Ticket Granting Server.	3	L2	1.1.1
7. A system administrator needs to securely access a remote Linux server over an untrusted network. Apply SSH to the scenario and explain how it provides secure remote access.	3	L3	2.1.4
8. Explain the structure of a JSON Web Token (JWT) and describe the OAuth 2.0 authorization flow.	4	L2	1.1.1
9. An organization uses Kerberos for centralized authentication. Apply the Kerberos authentication process to explain how a user obtains a ticket and accesses a protected service.	5	L3	2.1.4
10. A mobile application needs to access a user's resources from an external service without obtaining the user's password. Apply OAuth 2.0 and JWT-based authentication/authorization to this scenario and describe the token flow.	5	L3	2.1.4

Chapter wise Plan

Course Code and Title: **26ECAC305**Cryptography and Network Security	
Chapter Number and Title: 7. Network and Wireless Security Protocols	Planned Hours: **4 hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Sr.No	TLO's	CO's	BL	CA Code
1	Discuss the key elements of the 5G network security architecture	CO3	L2	1.1
2	Apply IPSec concepts to identify appropriate security mechanisms for different network communication scenarios.	CO3	L3	2.1
3	Apply IKE concepts to establish and manage secure communication associations.	CO3	L3	2.1
4	Differentiate and select appropriate wireless security mechanisms such as WPA2 and WPA3 for given network scenarios.	CO3	L3	2.1

Lesson Schedule Class No. - Portion covered per hour
1. IPSec overview, Encapsulating Security Payload,
2. Internet Key Exchange, Wireless security: IEEE 802.11, modern wireless security protocols (WPA2/WPA3).
3. 5G network security architecture
4. Secure communication in distributed systems

Review Questions

Sr.No. - Questions	TLO	BL	PI Code
1. Explain the IPSec architecture and describe the format and working of the Encapsulating Security Payload (ESP)	1	L2	1.1.1
2. Explain the purpose of Authentication Header (AH) and Encapsulating Security ayload (ESP) in IPSec. Compare their security services.	1	L2	1.1.1
3. Given two branch routers communicating over the public Internet through an ?Sec VPN, apply Internet Key Exchange (IKE) to authenticate the routers and stablish shared session keys for secure data transmission.	1	L2	2.1.3
4. An organization wants to establish a secure site-to-site VPN between its eadquarters and a remote branch office. Apply IPSec and explain how Security ssociations and IKE can be used to establish secure communication.	1	L3	2.1.3

5. Explain the security mechanisms used in IEEE 802.11 wireless LANs. Describe ıe role of authentication and encryption in protecting wireless communication.	2	L2	1.1.1
6. Compare WPA2 and WPA3 with respect to authentication, encryption, key stablishment, and protection against password-based attacks.	2	L2	1.1.1
7. A university provides Wi-Fi access to students, faculty, and visitors. Apply suitable EE 802.11 security mechanisms and WPA3 features to protect the wireless network om unauthorized access and data interception.	2	L3	2.1.4
8. Given a 5G network comprising user devices, base stations, and a core network, pply appropriate authentication and encryption mechanisms to protect ommunication against unauthorized access and data interception.	3	L3	2.1.4
9. Explain the major security challenges in distributed systems and describe suitable pproaches for establishing secure communication between distributed nodes.	4	L3	2.1.4
10. A distributed system consists of multiple nodes communicating over an untrusted etwork. Apply suitable authentication, encryption, and key-management mechanisms protect communication between the nodes.	4	L3	2.1.4
n organization uses an unsecured wireless LAN where attackers can capture ansmitted traffic. Analyze the security weakness and apply a suitable 802.11 security ıechanism to protect the network.	2	L3	2.1.4
2.A 5G network consists of user devices, base stations, and a 5G core network. pply appropriate 5G security mechanisms to protect communication between these omponents against unauthorized access and data interception.	3	L3	2.1.4

Chapter wise Plan

Course Code and Title: **26ECAC305** Cryptography and Network Security	
Chapter Number and Title: 8. **Emerging Cryptography**	Planned Hours: **4 hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Sr.No	TLO's	CO's	BL	CA Code
1	Explain the need for Post-Quantum Cryptography and identify the key characteristics of major PQC algorithms	4	L2	1.1
2	Compare classical cryptographic techniques with PQC and evaluate the role of PQC in enhancing network security.	4	L3	2.1
3	Identify the design requirements and constraints of lightweight cryptography in emerging networks.	4	L3	2.1
4	Analyze lightweight cryptographic algorithms for secure IoT and edge networks.	4	L3	2.1

Lesson Schedule Class No. - Portion covered per hour
1. Post-Quantum Cryptography: Post-Quantum Cryptographic (PQC) Algorithms,
2. Classical cryptography vs. PQC, PQC in Network Security,
3. Lightweight Cryptography in Emerging Networks: Design considerations
4. Lightweight cryptographic algorithms, Applications in IoT and edge networks.

Review Questions

Sr.No. - Questions	TLO	BL	PI Code
1.Explain how a suitable Post-Quantum Cryptographic (PQC) algorithm can be used to secure data transmission in a network vulnerable to quantum attacks	1	L2	2.1.4
2.Explain with an example how quantum computing affects the security of classical cryptographic algorithms	2	L2	2.1.4
3. Given an IoT device with limited power, memory, and computational resources, apply suitable design considerations for selecting a lightweight cryptographic algorithm.	3	L3	2.1.4

4. A smart agriculture IoT network uses soil-moisture and temperature sensors to transmit data wirelessly to an IoT gateway. The resource-constrained sensors require lightweight cryptography to ensure data confidentiality and integrity. Given this IoT network scenario, apply a suitable lightweight cryptographic technique to provide confidentiality and integrity for the sensor data transmitted from the sensors to the gateway.	4	L3	2.1.4
5.List and explain the security requirements of lightweight cryptography for resource-constrained devices.	3	L2	2.1.4
6.Given a network that uses RSA for secure key exchange, analyze the impact of quantum attacks and select an appropriate PQC-based alternative for the network.	2	L3	2.1.4
7.Multiple resource-constrained IoT devices communicate with an edge server. Apply lightweight cryptographic mechanisms to provide confidentiality, integrity, and authentication while minimizing computational overhead.	3	L3	2.1.4
8. A sensor node needs to provide confidentiality and integrity while operating with limited computational resources. Select and apply a suitable lightweight cryptographic algorithm for the sensor node and justify your choice.	2	L3	2.1.4
9. An organization wants to make its VPN infrastructure resistant to future quantum attacks. Apply PQC to the key-establishment process and explain how it strengthens network security. more generic way	3	L3	2.1.4

Model Question Paper for Minor Examination (ISA-1)					
Course Code: **26ECAC305**	Course Title: **Cryptography and Network Security**				
Duration: **1 Hr 15 minutes**	Max. Marks: **40**				
Q.No	Questions	Marks	CO	BL	PI Code
1a	Explain the OSI Security Architecture. Describe the major categories of security attacks, security services and security mechanisms.	6	1	L2	1.1.1
1b	Encrypt the message “COMPUTER SCIENCE” using the Playfair cipher with the keyword “SECURITY”. Show the construction of the key matrix and the encryption steps.	6	1	L3	2.1.3
1c	Describe various types of cryptanalytic attacks based on what is known to cryptanalyst.	6	1	L2	1.1.1
2a	Explain the structure and working of AES with a neat diagram. Describe the functions of the major AES transformations.	6	1	L2	1.1.1
2b	A system encrypts a message using AES but does not provide authentication. An attacker modifies some ciphertext bits before delivery. Identify the security problem and explain how authenticated encryption can prevent such modification.	6	1	L3	2.1.4
2c	Explain ECC Diffie Hellman key exchange algorithm. Consider elliptic curve y2 =x3+x+7 mod17 with point P= (2, 0) and Q= (1, 3). Find the following. i) P+Q ii) 2P	8	1	L2	1.1.1
3a	With a neat diagram explain one round of DES with an example.	6	1	L2	1.1.1
3b	A block cipher uses a 64-bit block size to encrypt a file of 150 bytes. Determine the number of complete blocks and the number of padding bytes required. Explain why padding is necessary.	6	1	L3	2.1.3
3C	Obtain the plaintext message for the cipher text “SA KN OX ” using the Hill cipher method. The enciphering matrix which used to obtain the above cipher text is $\begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$	8	1	L3	2.1.3

Model Question Paper for Minor Examination (ISA-2)	
Course Code: **26ECAC305**	Course Title: **Cryptography and Network Security**
Duration: **1 Hr 15 minutes**	Max. Marks: **40**
Note: Answer any two full questions. All questions carry equal marks	

Q.No	Questions	Marks	CO	BL	PI Code
1a	With a neat diagram, explain the major stages of the SHA-512 algorithm. Describe the role of the message schedule and compression function.	6	3	L2	1.1.1
1b	A software provider publishes the SHA-512 hash value of a software package. After downloading the package, a user obtains a different hash value. Apply the properties of cryptographic hash functions to determine what this indicates about the downloaded file.	6	3	L2	1.1.1
1c	A user wants to digitally sign a message using RSA. Given p = 5, q = 11 and e = 3, and message digest M = 9, calculate the RSA private key and show the digital signature generation and verification process.	8	3	L3	1.1.1
2a	Explain the Access Control Matrix and Role-Based Access Control (RBAC) models with suitable examples.	6	3	L2	1.1.1
2b	A company has three roles—Administrator, Faculty and Student—and three resources—Student Records, Course Materials and Examination System. Construct an appropriate Access Control Matrix specifying the permissions for each role.	6	3	L2	1.1.1
2c	A university has a centralized authentication system for students and faculty. Apply a suitable authentication protocol to this scenario and explain how a user can securely authenticate and access a protected service. Clearly describe the authentication flow.	8	3	L3	1.1.1
3a	Explain the SSL/TLS architecture and handshake procedure with a neat diagram.	6	4	L2	1.1.1
3b	A company wants to secure communication between a web browser and its online banking server. Apply SSL/TLS to the given scenario and describe the major steps involved in establishing a secure session.	6	4	L2	1.1.1
3c	A mobile application needs to access a user's resources from an external service without obtaining the user's password. Apply OAuth 2.0 and JWT-based authorization to this scenario and explain the complete token flow.	8	4	L3	1.1.1

Model Question Paper – ESA

Question Paper Title: Model Question Paper for ESA					
Course: Cryptography and Network Security		**Course Code: 26ECAC305**			
Total Duration: 3 Hrs		**Maximum Marks: 100**			
Note: Answer any 5 full questions, at least one from each unit.					
Q.No	**Questions**	**Marks**	**CO**	**BL**	**PI Code**
	Unit I				
1a	Explain the major categories of security attacks, security services, and security mechanisms	6	1	L2	1.1.1
1b	A college network observes that an attacker is secretly reading students' login credentials while they are being transmitted. Identify the attack, identify the security property affected, and suggest one suitable security mechanism to protect the communication.	6	1	L3	1.1.1
1c	Alice and Bob use the Diffie-Hellman key exchange algorithm with q = 23 and α = 5. Alice selects private key XA = 6 and Bob selects private key XB = 15. Calculate Alice's public key, Bob's public key, and the common secret key. Show all necessary calculations.	8	1	L3	2.1.3
2a	With a neat diagram, explain the structure and working of AES. Describe the functions of SubBytes, ShiftRows, MixColumns and AddRoundKey operations.	6	1	L2	1.1.1
2b	A block cipher uses a 64-bit block size to encrypt a file of 150 bytes. Determine the number of complete blocks and the number of padding bytes required. Explain why padding is necessary.	6	1	L3	2.1.3
2c	In an RSA system, let p = 5, q = 11 and e = 3. Calculate n, φ(n) and the private key d. Encrypt the message M = 9 and show the necessary modular exponentiation steps.	8	1	L3	2.1.3
	Unit II				
	Explain the requirements of a cryptographic hash function. Discuss the importance of collision resistance in maintaining data integrity.	6	2	L2	1.1.1
3b	A software provider publishes the SHA-512 hash value of a software package. After downloading the package, a user obtains a different hash value. Apply the properties of cryptographic hash functions to determine what this indicates about the downloaded file.	6	2	L3	1.1.1
3c	A user wants to digitally sign a message using RSA. Given p = 5, q = 11, e = 3 and message digest M = 9, calculate the RSA private key and demonstrate the digital signature generation and verification process.	8	2	L3	1.1.1

4a	Explain the different authentication methods, including passwords, Two-Factor Authentication (2FA) and Multi-Factor Authentication (MFA).	6	2	L2	2.1.3
4b	A university provides students access to an online examination portal. Apply an appropriate authentication method using passwords and 2FA, and describe the authentication process.	6	2	L3	2.1.4
4c	A company has three roles: Administrator, Faculty and Student and three resources: Student Records, Course Materials and Examination System. Construct an Access Control Matrix specifying suitable access permissions for each role. Explain how Role-Based Access Control (RBAC) can be applied to implement these permissions.	8	2	L3	2.1.4
Unit III					
5a	With a neat diagram, explain the SSL/TLS handshake procedure and describe the purpose of the major messages exchanged during the handshake.	6	3	L2	1.1.1
5b	A company wants to secure communication between a web browser and its online banking server. Apply the SSL/TLS protocol to this scenario and describe the major steps involved in establishing a secure session.	6	3	L3	2.1.4
5c	A mobile application needs to access a user's resources from an external service without obtaining the user's password. Apply OAuth 2.0 and JWT-based authentication/authorization to this scenario and describe the complete token flow.	8	3	L3	2.1.4
6a	Explain the working of the Kerberos authentication system. Describe the roles of the Authentication Server (AS) and Ticket Granting Server (TGS).	6	3	L2	1.1.1
6b	A system administrator needs to securely access a remote Linux server over an untrusted network. Apply SSH to the scenario and explain how it provides secure remote access.	6	3	L3	2.1.3
6c	A smart agriculture IoT network uses soil-moisture and temperature sensors to transmit data wirelessly to an IoT gateway. The resource-constrained sensors require confidentiality and integrity for the transmitted data. Apply a suitable lightweight cryptographic technique to secure the sensor data and justify your choice based on computational capability, memory usage, power consumption and security strength.	8	4	L3	2.1.4